Time Series Anomaly Detection: Methods, Scores, and Thresholds

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Anomaly is not just noise in the data

In **outlier analysis**, the extreme value is typically treated as noise to be removed.

In **anomaly detection**, the atypical value is the **object of interest**.

Examples where the anomaly carries information:

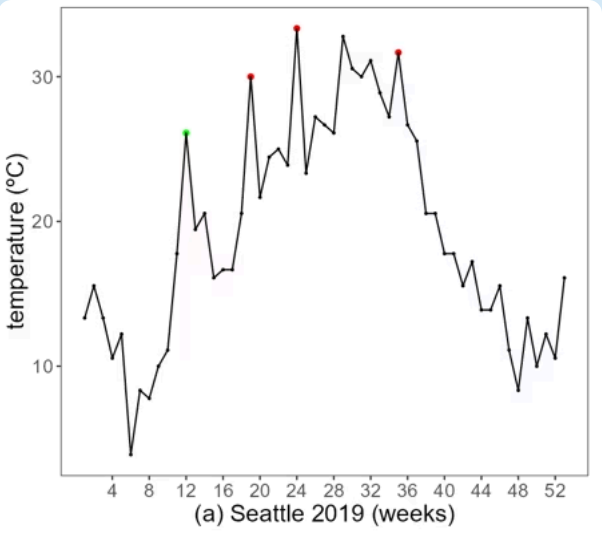
- Fraud
- System failures
- Epidemic outbreaks
- Security incidents
- Rare events

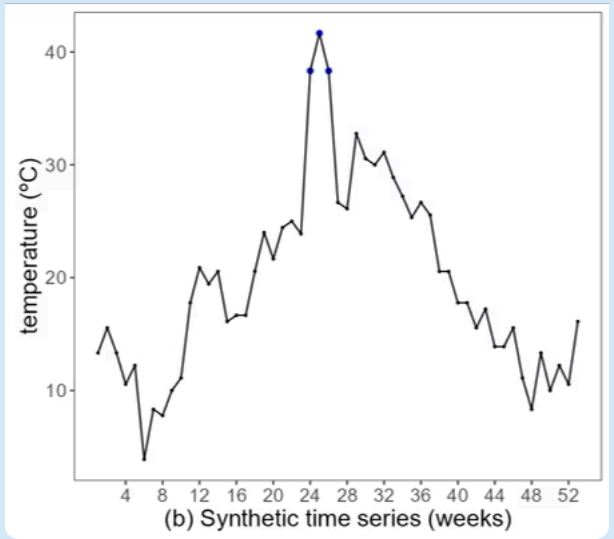
> Minimal formalization: an anomaly is a rare event $a \in S$ such that $P(a)$ is low and the information of a is high — that is, $I(a) = -\log P(a) \gg 0$.

Anomaly Taxonomy

- **Type:** point or sequential
- **Circumstance:** contextual or out-of-context
- **Dimension:** univariate or multivariate
- **Interpretation:** supervised, semi-supervised, or unsupervised
- **Form:** how the detector operates (regression, classification, clustering...)

The taxonomy organizes type, circumstance, dimension, interpretation, and learning form.





Point and Sequential Anomalies

Point Anomalies

- A single observation differs significantly from the rest of the series.
- The decision depends on the local context and the model of typical behavior.
- It is the most straightforward form of anomaly, but not necessarily the easiest to detect.

Sequential Anomalies

- An entire subsequence exhibits unusual behavior.
- The isolated point may seem acceptable, but the complete sequence reveals the event.
- Requires comparing windows, shapes, or temporal representations.

Example with weekly temperature: point anomaly in panel (a) and anomalous sequence in panel (b).

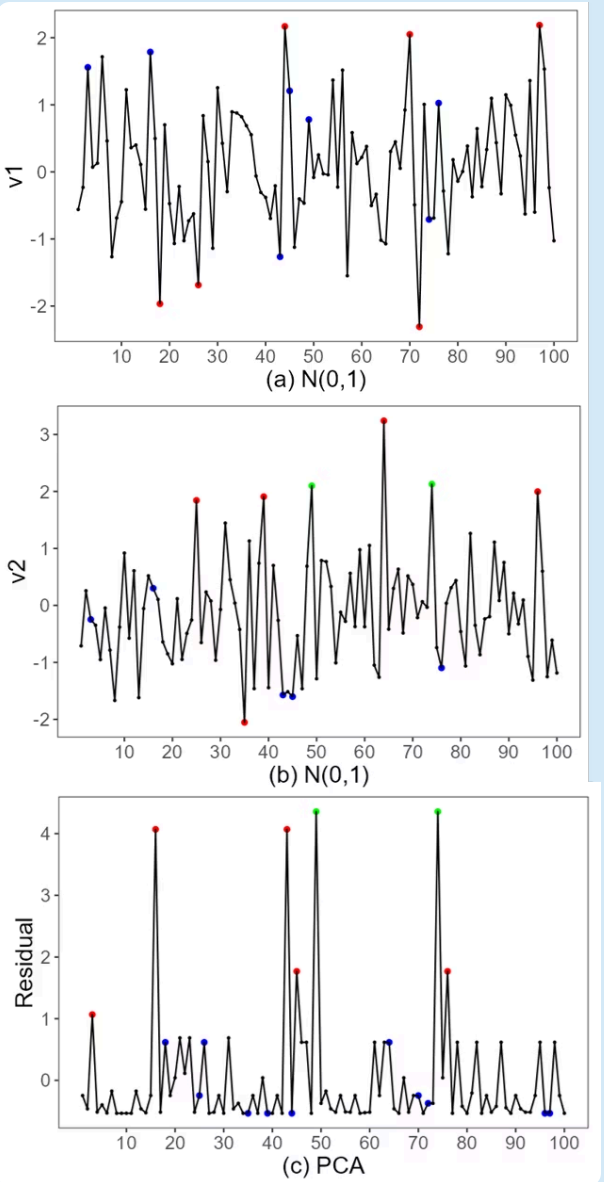
Contextual and Out-of-Context Anomalies

Contextual

- The observation is not strange on its own.
- It becomes anomalous when considered in its temporal context.
- Example: 26°C may be common in summer, but anomalous in winter.

Out of Context

- The value is atypical regardless of when it occurs.
- The explanation does not strongly depend on the temporal neighborhood.
- This case approximates the classical notion of a statistical outlier.



Univariate vs multivariate

Univariate

A single variable is analyzed over time.

The anomaly appears as an atypical value or sequence in that series.

Methods are simpler and more interpretable.

Multivariate

The anomaly may arise from the combination of variables.

Each variable in isolation may appear normal.

Relationships between dimensions become important.

Example: multivariate anomaly with PCA

The example shows anomalies captured by the relationship between variables, not just by individual extreme values.

Labeled and Unlabeled Series

Labeled

Each observation can indicate the presence or absence of an anomaly.

Enables evaluation and supervised learning.

Labels may be incomplete or temporally imprecise.

Unlabeled

There is no direct indication of what constitutes an anomaly.

Detection relies on statistical hypothesis, distance, or rarity.

This is the most common case in real-world applications.

Learning Scenarios

Supervised

- Uses observations labeled as typical/anomalous.
- The anomaly is treated as a class to be predicted.
- The central challenge is the scarcity and imperfection of labels.

Semi-supervised

- Trains the model using only typical data.
- New observations are considered anomalous when they do not fit the learned pattern.
- Common when the anomalous class is rare or difficult to label.

Unsupervised

- Does not require labels for training.
- Assumes that typical observations are more frequent or denser.
- Flexible, but requires greater care in interpreting alerts.

When to use each learning scenario

More labels

- Supervised when labels are reliable.
- Useful in online operation after a training phase.
- Allows optimizing metrics such as F1.

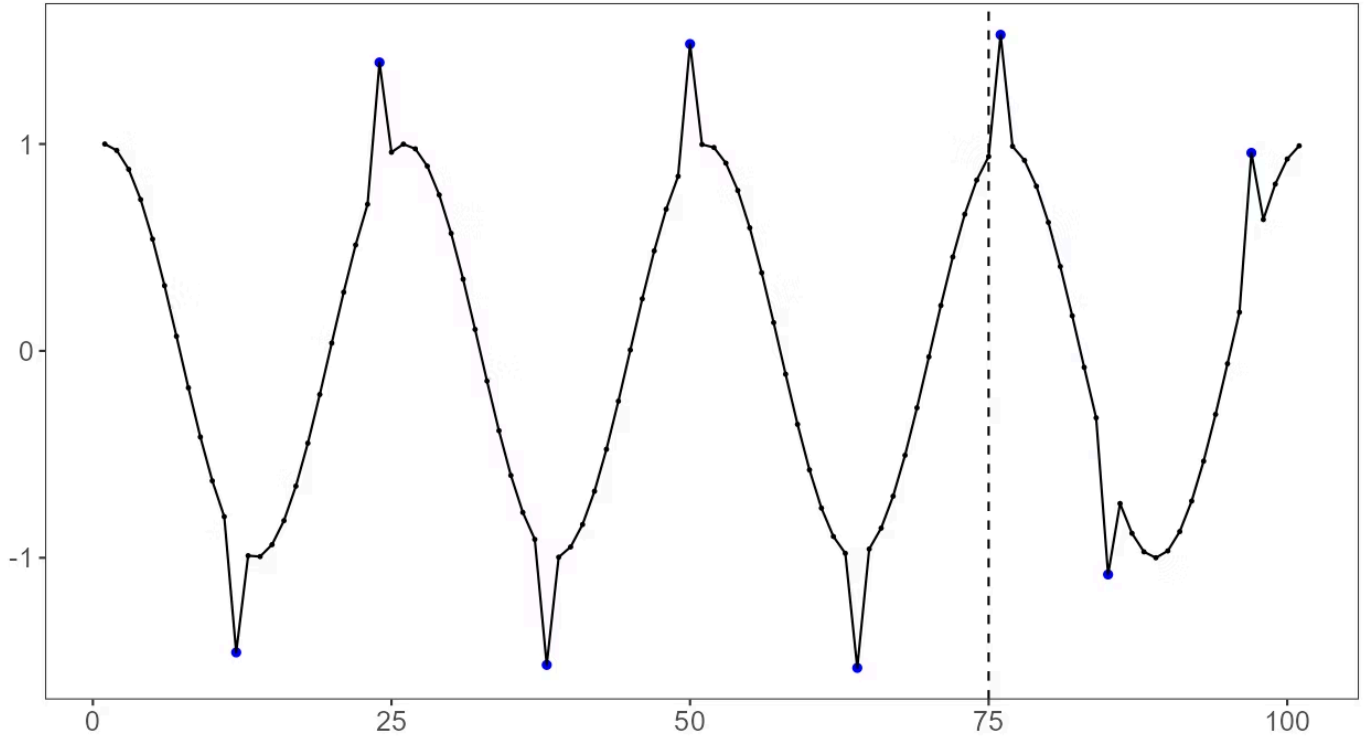
Fewer labels

- Semi-supervised when there are reliable typical examples.
- Unsupervised when there is almost no label.
- Requires subsequent analysis of alarms.

Baseline Example for Comparing Methods

The same series is used to contrast families of detectors.

The goal is to understand the operational hypothesis of each family, not to rank algorithms.



Harbinger synthetic series: training and test regions allow comparison of detector families.

Sliding Window: Transforming Time into Features

A **window of size p** uses past observations as features.

The **target** is the next observation in the series.

At each time step, the window advances one position and produces another training or test example.

> $x_{t-p}, \ldots, x_{t-2}, x_{t-1} \quad x_t$

```
INPUT: series X = x1, ..., xn , window size p
PRECONDITION: n > p

STEP 1 — For t from p+1 to n:
  features ← xt−p, ..., xt−1
  target ← xt
  add (features, target) to dataset D
STEP 2 — Split D into:
  Dtrain (earlier time steps)
  Dtest  (later time steps)
  preserving temporal order

OUTPUT: Dtrain and Dtest ready
        for supervised regression
POSTCONDITION: no future data leaks into training
```

Anomaly Detection by Regression

Core idea

A model learns the expected behavior of the series. An anomaly is the deviation between the observed xt and the estimated x̂t.

> ωt = distance(xt, x̂t)

Sliding window

Past observations ⟨xt−p, ..., xt−1⟩ form the attributes; xt is the target. At each time step the window advances one position.

> [xt−4, xt−3, xt−2, xt−1] → xt

Tabular example

With p = 5, one row of the training set contains four lags and one target. Care must be taken to preserve temporal order: training first, testing after.

The lagged terms xt−4,...,xt−1 predict xt; the difference between prediction and observation becomes the residual.



t	x_{t-4}	x_{t-3}	x_{t-2}	x_{t-1}	$\hat{x}_t$	x_t
5	v_1	v_2	v_3	v_4	$\hat{v}_5$	v_5
6	v_2	v_3	v_4	v_5	$\hat{v}_6$	v_6
7	v_3	v_4	v_5	v_6	$\hat{v}_7$	v_7
8	v_4	v_5	v_6	v_7	$\hat{v}_8$	v_8
9	v_5	v_6	v_7	v_8	$\hat{v}_9$	v_9
10	v_6	v_7	v_8	v_9	$\hat{v}_{10}$	v_{10}
11	v_7	v_8	v_9	v_{10}	$\hat{v}_{11}$	v_{11}
12	v_8	v_9	v_{10}	v_{11}	$\hat{v}_{12}$	v_{12}
13	v_9	v_{10}	v_{11}	v_{12}	$\hat{v}_{13}$	v_{13}
14	v_{10}	v_{11}	v_{12}	v_{13}	$\hat{v}_{14}$	v_{14}

(a)

From Error to Anomaly Score

After prediction, a distance between xt and x̂t is computed.

This value is the model's **deviation score**.

The higher the score, the greater the evidence that the observed behavior was not expected by the model.

> **ωt = distance(xt, x̂t)**

```
INPUT: series X = x1, ..., xn , trained model M,
      distance function distance(·,·)
PRE-CONDITION: M was trained only on typical data

STEP 1 — For each time instant t in the test set:
  x̂t ← M.predict( xt−p, ..., xt−1 )
  ωt ← distance(xt, x̂t)
STEP 2 — Build score series
  W = ω1, ..., ωn
STEP 3 — Compute threshold τ from W
  (gaussian, boxplot or EVT)
STEP 4 — Mark t as anomalous if ωt > τ

OUTPUT: set of anomalous instants A ⊆ {1, ..., n}
POST-CONDITION: A contains only test instants
```

Distance choices

Absolute error

- Preserves the original scale of the series.
- Easy to explain to domain experts.
- Less aggressive with very large errors.

Squared error

- Amplifies large errors.
- Common in regression and optimization.
- Can increase sensitivity to spikes.

How to decide the threshold

The model generates a series of residuals W = ⟨ω1, ..., ωn⟩. Detection depends on separating typical residuals from extreme residuals.

Gaussian threshold

- Assumes that typical residuals approximately follow a Gaussian distribution.
- Simple, but fragile when the distribution is asymmetric, multimodal, or non-stationary.

$$\omega_t > \text{mean}(W) + 3 \cdot \text{sd}(W)$$

Boxplot threshold

- Does not require assuming a specific distribution for the residuals.
- Uses quartiles and interquartile range.
- In the case of regression, the upper bound of errors is typically of interest.

$$\omega_t > Q3(W) + 1{,}5 \cdot IQR(W)$$

Extreme value theory

- Specifically models the tail of the error distribution.
- Useful when relevant events are rare and extreme.
- The threshold u is chosen using the POT (Peaks Over Threshold) method.

$$P(\omega > u + y \mid \omega > u) \approx \left(1 + \frac{\xi \cdot y}{\beta}\right)^{-1/\xi}$$

u: exceedance threshold; ξ: tail index; β: scale

Trend Followers

Follower models smooth or decompose the series to estimate the expected behavior.

Examples: moving average, splines, FFT, wavelet, EMD, REMD, and FBIAD.

The anomaly appears when the observation deviates from the curve followed by the model.

> **General formula:** An anomaly is detected when
>
> $$|x_t - \hat{f}(t)| > \tau$$
>
> where $\hat{f}(t)$ is the curve estimated by the follower and τ is the detection threshold.

- **Limitation:** Smooth followers may not capture short-duration anomalies, since smoothing attenuates point deviations before they exceed the threshold.

Statistical Regression Models

ARIMA and linear regression model the series with explicit statistical assumptions.

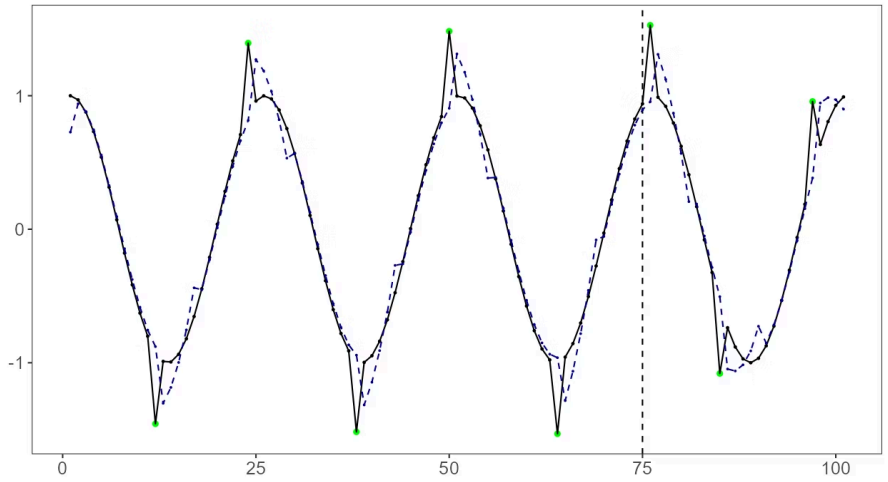
They work well when there is interpretable temporal structure, such as autocorrelation, trend, or seasonality.

Reading the residual is usually straightforward: what is left over after the model.

> **Distinction between models:**
>
> **ARIMA:** autoregressive and moving average structure — residual ε_t is white noise.
>
> **Linear regression:** $x_t = \beta_0 + \beta_1 \cdot t + \ldots + \varepsilon_t$ — residual ε_t is the deviation from the trend.

- **Limitation:** Both assume stationarity or linear trend; regime changes may generate false positives.

ARIMA as a deviation-based detector



The fitted model estimates the series; high residuals around anomalies support detection.

Anomaly: $|\varepsilon_t| > \tau$

INPUT: series X = x1, ..., xn , orders (p,d,q), threshold τ
PRE-CONDITION: series stationary after d differentiations

STEP 1 — Fit ARIMA(p,d,q) on X_train
STEP 2 — For each time step t in test:
  x̂t ← ARIMA.predict(t)
  εt ← xt − x̂t
STEP 3 — Build W = ε1, ..., εn
STEP 4 — Compute τ
  (gaussian: mean(W) + 3·sd(W))
STEP 5 — Flag t as anomalous if |εt| > τ

OUTPUT: anomalous time steps A
POST-CONDITION: training residuals must be white noise

Machine Learning Regressors

The sliding window allows the use of general supervised models.

Examples: KNN, SVM, Random Forest, XGBoost, MLP, and ELM.

The detector remains based on the error between prediction and observation.

> **KNN and SVM:** distance/margin-based — good for local patterns.
>
> **Random Forest and XGBoost:** tree ensembles — capture non-linearities.
>
> **MLP and ELM:** shallow neural networks — universal approximators.

⚠ **Caution:** All depend on the sliding window to transform the series into tabular features.

Deep Regressors

Conv1D and LSTM exploit temporal patterns in sequences.

DeepAnT uses convolutional networks to predict the next window and detect deviations.

Telemanom and LSTM-based detectors use prediction error to locate unexpected behavior.

⚠️ **Limitation:** They require more training data and are harder to interpret than statistical models.

Conv1D: captures local patterns via convolutional filters.

LSTM: captures long-term dependencies via recurrent memory.

DeepAnT: uses Conv1D to predict the next window and compares it with the observed value.

Recent and Hybrid Methods

- Triple Exponential Smoothing incorporates level, trend, and seasonality.
- Median Method uses the median of the contextual window as a robust forecast.
- Hybrid methods and ensembles combine models to reduce individual weaknesses.
- Transformers and foundation models (e.g., Anomaly Transformer, PatchTST) apply attention mechanisms to capture long-range dependencies.

⚠ More complex methods do not guarantee better detection — the choice should consider interpretability, computational cost, and availability of labeled data.

What changes between regression types

The expected model changes

ARIMA, moving average, spline, RF, XGBoost, LSTM.

Each one captures a different form of temporal pattern.

A bad model generates a bad residual.

The logic remains

Predict xt.

Calculate the error.

Compare the error against a normality reference.

The threshold changes

Gaussian (parametric): assumes normal distribution of residuals.

Boxplot (non-parametric): based on quartiles, with no distribution assumption.

EVT (tail): models rare extremes using extreme value theory.

Practical Considerations

- The window p controls how much temporal memory the model receives.
- Training with anomalies can cause the model to learn the event as if it were normal.
- A regime change may look like an anomaly to a non-adaptive model.
- Normalization and scaling directly affect the error.
- Temporal validation: never use future data to estimate the threshold — the threshold must be calculated only on the training set.

> The quality of the detector depends as much on the forecasting model as on the thresholding strategy and the integrity of the train/test split.

Classification: anomaly as a class

- A classifier learns to separate typical and anomalous observations.
- During training, labels must be available.
- During testing, the model predicts the label of new observations.
- Class imbalance is the main challenge — anomalies are rare by definition.

> **Binary classification:** typical (0) vs anomalous (1).
> **Multiclass classification:** type of anomaly as a class.
> **One-class classification:** trains only on typical data (e.g., One-Class SVM).

Sliding window

In the training stage, the window has a label; in the testing stage, the model predicts the presence of an anomaly.



t	x_{t-4}	x_{t-3}	x_{t-2}	x_{t-1}	x_t	$\hat{e}_t$	e_t
5	v_1	v_2	v_3	v_4	v_5	$\hat{b}_5$	b_5
6	v_2	v_3	v_4	v_5	v_6	$\hat{b}_6$	b_6
7	v_3	v_4	v_5	v_6	v_7	$\hat{b}_7$	b_7
8	v_4	v_5	v_6	v_7	v_8	$\hat{b}_8$	b_8
9	v_5	v_6	v_7	v_8	v_9	$\hat{b}_9$	b_9
10	v_6	v_7	v_8	v_9	v_{10}	$\hat{b}_{10}$	b_{10}
11	v_7	v_8	v_9	v_{10}	v_{11}	$\hat{b}_{11}$	b_{11}
12	v_8	v_9	v_{10}	v_{11}	v_{12}	$\hat{b}_{12}$	b_{12}

(a)

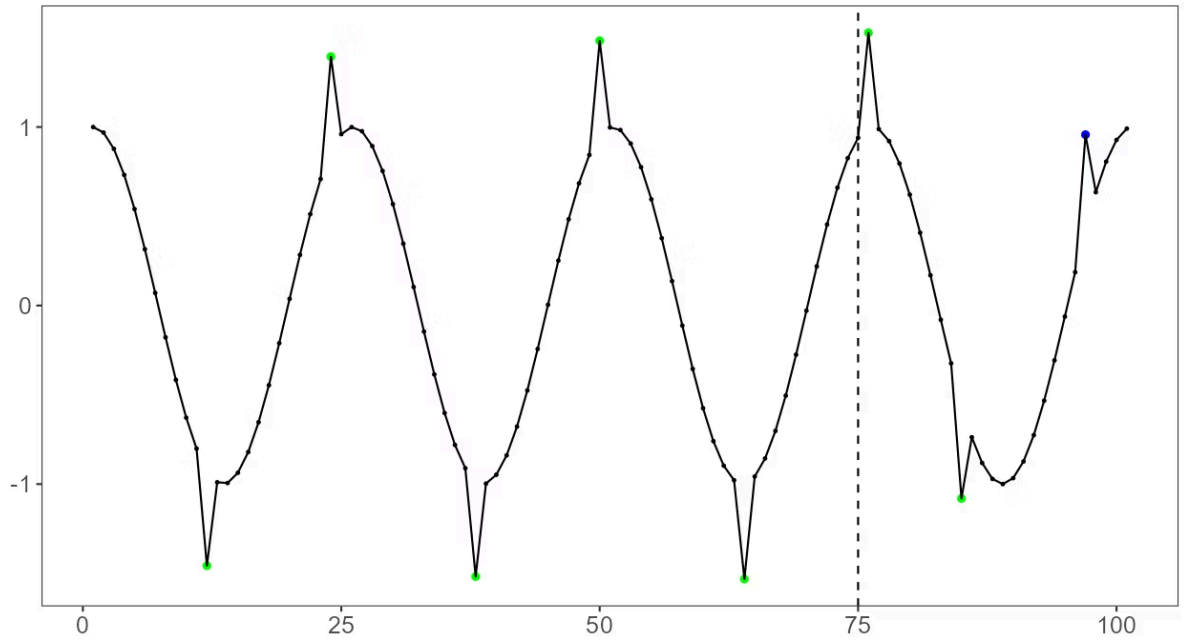
t	x_{t-4}	x_{t-3}	x_{t-2}	x_{t-1}	x_t	$\hat{e}_t$
13	v_9	v_{10}	v_{11}	v_{12}	v_{13}	$\hat{b}_{13}$
14	v_{10}	v_{11}	v_{12}	v_{13}	v_{14}	$\hat{b}_{14}$

(b)

Example with SVM

Classification challenge

- Anomalies are rare, creating strong class imbalance.
- Human labels may miss events or mark the moment with a delay.
- The learned class may not represent future anomalies not yet seen.



The SVM detects some of the anomalies in the test; the example also highlights a false negative.

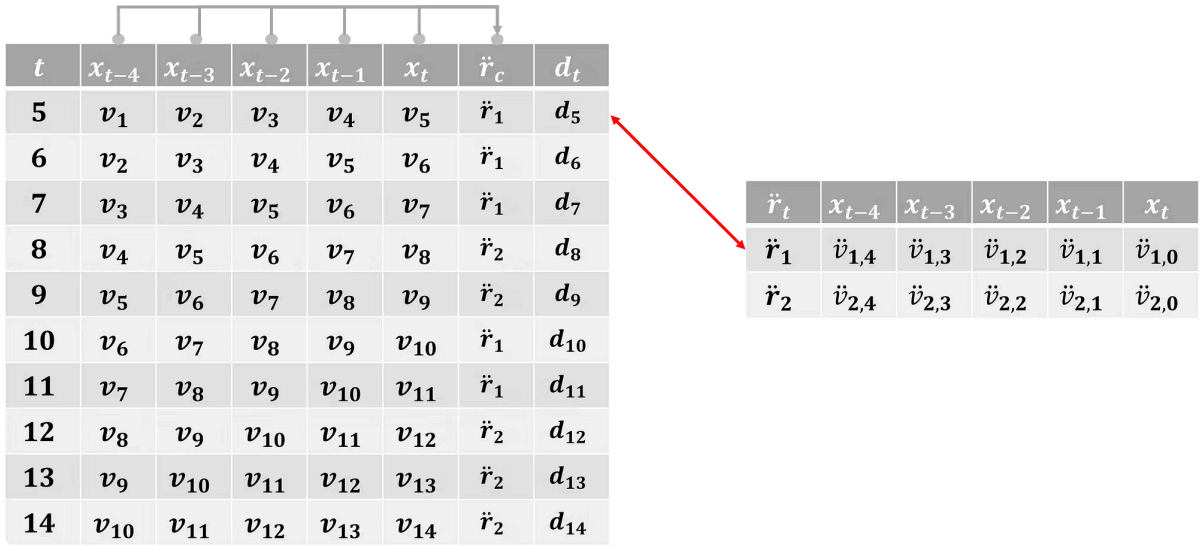
Clustering: Anomaly as Distance

- Typical observations form groups or dense regions.
- Anomalies appear far from representatives or outside the groups.
- The approach is naturally unsupervised.
- Examples: K-Means (centroids), DBSCAN (density), LOF (local outlier factor).

General anomaly score formula:

$$\text{score}(x_t) = \min_k \text{dist}(x_t, c_k)$$

where *ck* are the cluster representatives (centroids).



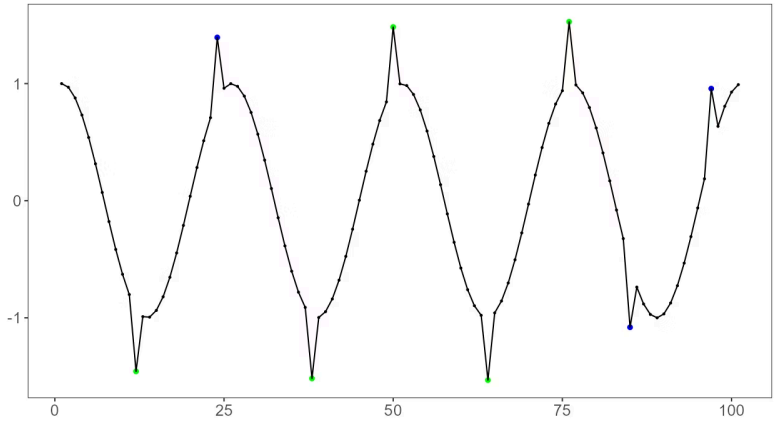
t	x_{t-4}	x_{t-3}	x_{t-2}	x_{t-1}	x_t	$\ddot{r}_c$	d_t
5	v_1	v_2	v_3	v_4	v_5	$\ddot{r}_1$	d_5
6	v_2	v_3	v_4	v_5	v_6	$\ddot{r}_1$	d_6
7	v_3	v_4	v_5	v_6	v_7	$\ddot{r}_1$	d_7
8	v_4	v_5	v_6	v_7	v_8	$\ddot{r}_2$	d_8
9	v_5	v_6	v_7	v_8	v_9	$\ddot{r}_2$	d_9
10	v_6	v_7	v_8	v_9	v_{10}	$\ddot{r}_1$	d_{10}
11	v_7	v_8	v_9	v_{10}	v_{11}	$\ddot{r}_1$	d_{11}
12	v_8	v_9	v_{10}	v_{11}	v_{12}	$\ddot{r}_2$	d_{12}
13	v_9	v_{10}	v_{11}	v_{12}	v_{13}	$\ddot{r}_2$	d_{13}
14	v_{10}	v_{11}	v_{12}	v_{13}	v_{14}	$\ddot{r}_2$	d_{14}

$\ddot{r}_t$	x_{t-4}	x_{t-3}	x_{t-2}	x_{t-1}	x_t
$\ddot{r}_1$	$\ddot{v}_{1,4}$	$\ddot{v}_{1,3}$	$\ddot{v}_{1,2}$	$\ddot{v}_{1,1}$	$\ddot{v}_{1,0}$
$\ddot{r}_2$	$\ddot{v}_{2,4}$	$\ddot{v}_{2,3}$	$\ddot{v}_{2,2}$	$\ddot{v}_{2,1}$	$\ddot{v}_{2,0}$

Sliding window

Each window is associated with a group; the distance to the representative supports detection.

Example with K-Means



Sequences far from centroids are flagged as anomalies.

$$\text{score}(x_t) = \min_k \text{dist}(x_t, c_k)$$

INPUT: series X = x1, ..., xn , number of clusters k, threshold τ
PRE-CONDITION: sliding window already applied;
Xtrain represents typical behavior

STEP 1 — Apply K-Means(k) on Xtrain
→ centroids C = {c1, ..., ck}
STEP 2 — For each time instant t in test:
score(xt) ← min_{j=1..k} dist(xt, cj)
STEP 3 — Compute τ from training scores
(e.g.: 95th percentile)
STEP 4 — Flag xt as anomalous if score(xt) > τ

OUTPUT: anomalous instants A
POST-CONDITION: k and the distance metric must be validated in the domain

Limitations of Clustering

- The choice of k and the distance metric directly affects detection.
- Scaling and normalization can change which points appear distant.
- Data with anomalies during training may form spurious clusters.

Statistical Methods

- Assume that typical observations occupy regions of high probability.
- Anomalies appear in regions of low probability.
- Can use parametric models, hypothesis testing, or non-parametric profiles.

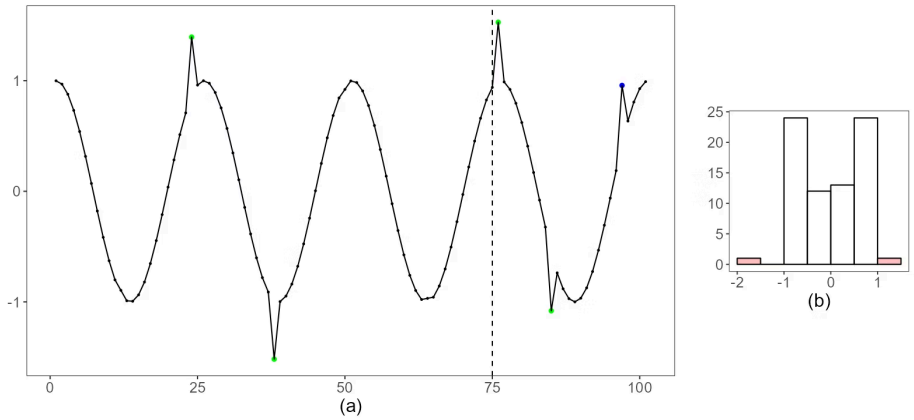
- Examples: histogram (non-parametric), KDE (kernel density estimation), Gaussian distribution (parametric), mixture of Gaussians (GMM).

General formula for the anomaly score:

$$\text{score}(x_t) = -\log P(x_t \mid \theta)$$

where θ are the parameters of the statistical model.

Example with Histogram



Rare bins → low frequency → high anomaly score

$$score(x_t) = \frac{1}{freq(bin(x_t))}$$

INPUT: series X = x1, ..., xn , number of bins B, threshold τ
PRECONDITION: Xtrain represents typical behavior

STEP 1 — Build histogram H with B bins over Xtrain
STEP 2 — For each bin b:
freq(b) ← count(b) / |Xtrain|
STEP 3 — For each time step t in test:
b ← bin(xt)
score(xt) ← 1 / freq(b) [or ∞ if freq(b) = 0]
STEP 4 — Mark xt as anomalous if score(xt) > τ

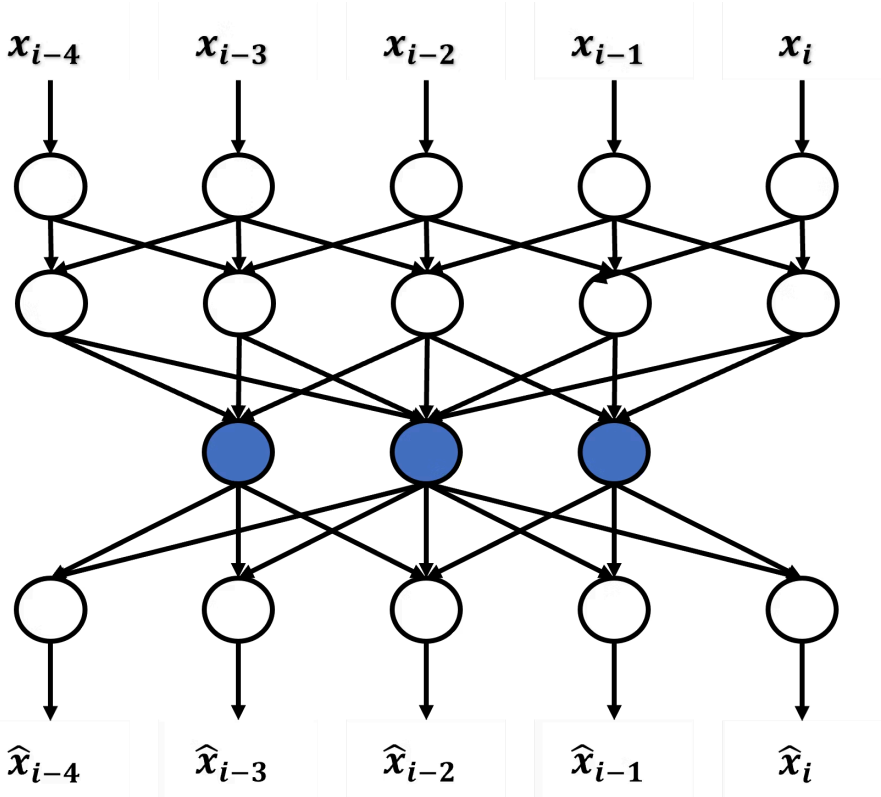
OUTPUT: anomalous time steps A
POSTCONDITION: bins with freq = 0 are always anomalous

Spectral Methods

- Project data into a lower-dimensional space.
- The main structure of the series must be preserved in the projection.
- Anomalies appear as observations that are poorly represented or poorly reconstructed.
- Examples: PCA, SVD, linear autoencoder — each defines a different form of projection and reconstruction.

> Reconstruction error as anomaly score:
>
> $$score(x_t) = \|x_t - P \cdot P^\top \cdot x_t\|^2$$
>
> where **P** are the retained principal components.

Autoencoder: architecture



Training loss function:

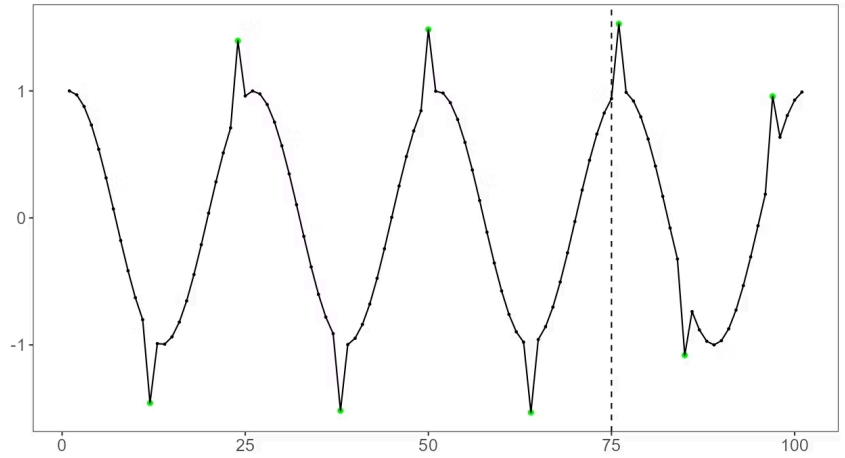
$$L = \|x - AE(x)\|^2 = \|x - D(E(x))\|^2$$

> where **E** is the encoder and **D** is the decoder.

- The encoder compresses x into a latent representation $z = E(x)$ of lower dimension.
- The decoder reconstructs $\hat{x} = D(z)$; the difference $\|x - \hat{x}\|^2$ is the reconstruction error.

The hourglass shape forces the network to learn a compact representation of typical patterns.

Autoencoder: reconstruction error



Poorly reconstructed sequences produce high error and are flagged as anomalous.

$$score(x_t) = \|x_t - AE(x_t)\|^2$$

INPUT: series X = x1, ..., xn , window p, threshold τ

PRECONDITION: AE trained only on typical data (Xtrain)

STEP 1 — Train AE on Xtrain minimizing $||x - AE(x)||^2$

STEP 2 — For each window xt = xt−p, ..., xt in the test set:

$\hat{x}t \leftarrow AE(xt)$

$score(xt) \leftarrow ||xt - \hat{x}t||^2$

STEP 3 — Compute τ from training scores

(e.g.: 95th percentile or Gaussian)

STEP 4 — Flag xt as anomalous if score(xt) > τ

OUTPUT: anomalous timestamps A

POSTCONDITION: AE must not have seen anomalies during training

Information Theory

Anomalies are rare and increase the information or entropy of the sample.

The question becomes: which subset changes the information of the series the most?

Compression and Kolmogorov complexity can serve as practical approximations.

> find $A \subset S$ that maximizes $| I(S) - I(S - A) |$

⚠️ Limitation

- Kolmogorov complexity is uncomputable; in practice, compression (`gzip`, `LZ77`) is used as an approximation.

✅ Application

- Useful for detecting sequential anomalies where the internal structure of the subsequence changes.

Volatility Anomalies

The anomaly is not necessarily in the level of the series, but in its variability.

ARCH and GARCH model instantaneous volatility.

This case is central in financial risk and in series with non-linear variation.

> **Level anomaly:** x_t deviates from the expected mean.
>
> **Volatility anomaly:** σ_t deviates from the expected variance pattern.

- ARCH models variance as a function of past residuals; GARCH adds past variance as a predictor.

Anomalous volatility: formalization

The temporal characteristic becomes the volatility v(xt).

The event occurs when the volatility deviates from what is expected before and after t.

$$va(X,k,\sigma) = \{t : |v(xt)-ep(v(xt),k)|>\sigma \wedge |v(xt)-ef(v(xt),k)|>\sigma\}$$

Parameters

k

Size of the past and future context window

σ

Deviation threshold to characterize the anomaly

ep(v(xt),k)

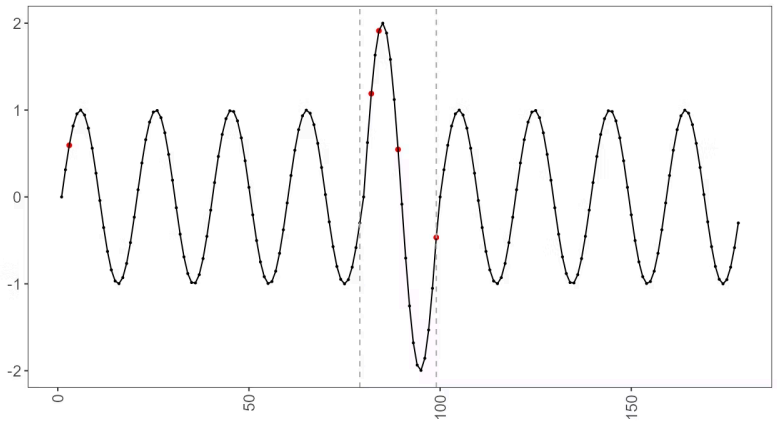
Expected volatility in the *k* instants prior to t

ef(v(xt),k)

Expected volatility in the *k* instants after t

- The bilateral condition (past **AND** future) reduces false positives in regime transitions.

Example with GARCH



Volatility peaks outside the GARCH confidence interval are flagged as anomalous.

GARCH Model:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \cdot \varepsilon_{t-1}^2 + \beta_1 \cdot \sigma_{t-1}^2$$

Anomaly: $|\varepsilon_t| > k \cdot \sigma_t$

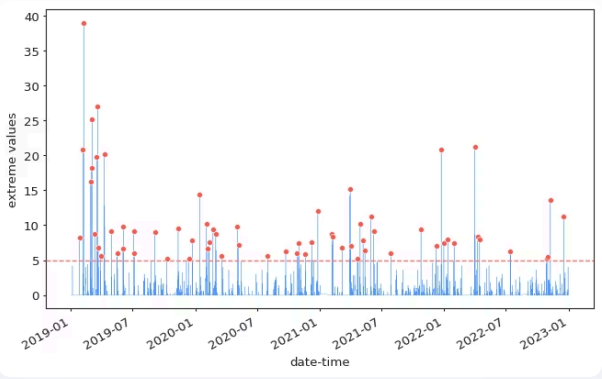
INPUT: series X = x1, ..., xn , GARCH orders (p,q), factor k
PRE-CONDITION: series with conditional heteroskedasticity
(significant ARCH test)

STEP 1 — Fit GARCH(p,q) on X_train
→ parameters α, β
STEP 2 — For each time step t in the test set:
$\sigma^2 t \leftarrow \alpha 0 + \Sigma\, \alpha i \cdot \varepsilon^2 t{-}i + \Sigma\, \beta j \cdot \sigma^2 t{-}j$
$\varepsilon t \leftarrow xt - \mu t$ [mean residual]
STEP 3 — Compute normalized volatility:
$zt \leftarrow \varepsilon t \,/\, \sigma t$
STEP 4 — Flag t as anomalous if $|zt| > k$
[e.g.: k = 3]

OUTPUT: anomalous time steps A with extreme volatility

POST-CONDITION: standardized residuals zt must be
i.i.d. on the training set

Extreme Values



Extreme values are rare, but relevant for decision-making.

The full distribution may be inadequate for modeling only the tail.

Extreme value theory treats extremes as a process of their own.

> **Mn = max{x1, x2, ..., xn}**

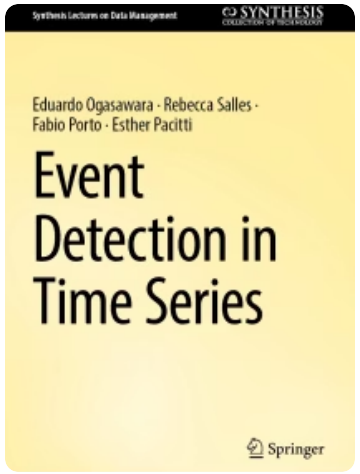
Fisher-Tippett-Gnedenko Theorem:

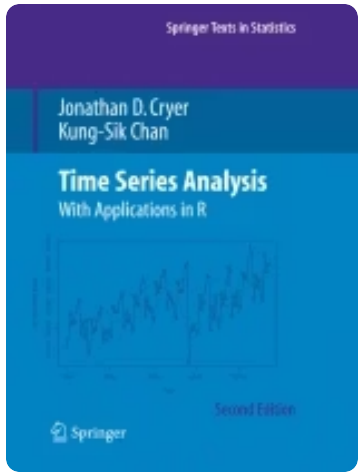
- The normalized maximum M_n converges to one of three extreme value distributions: **Gumbel** ($\xi = 0$), **Fréchet** ($\xi > 0$) or **Weibull** ($\xi < 0$), depending on the tail of the original distribution.
- In practice, the **GEV (Generalized Extreme Value)** distribution is used, which unifies all three cases.

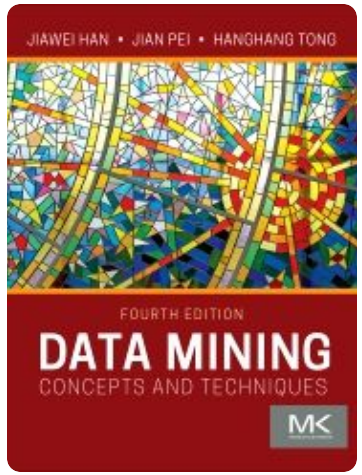
Example:

Extraction of extreme values using Peak over Threshold on precipitation data from 2019 to 2022.

Bibliographic References







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